

Individual reflection

In developing my model, my rationale was to move beyond a rudimentary application of logistic regression and, instead, build a model that was sensitive to the structure of the data, the class imbalance present and the clinical nature of the outcome. The initial logistic regression model served as a baseline for its interpretability and ubiquity in clinical prediction modelling [1]. However, the diagnostics revealed poignant deviations from linearity, particularly concerning renal function (serum creatinine) and cardiac markers (ejection fraction and CPK levels). Given that the biological processes underlying mortality risk are rarely linear, the decision to extend the model using natural cubic splines emerged as a way to ease restrictive linear assumptions while maintaining a level interpretability. This aligns with statistical literature emphasising flexible modelling of clinical risk functions [2].

Given the assessment rubric, the chosen performance metric was of high importance. For general discrimination, I used the ROC-AUC, a threshold-independent measure widely used in medical contexts [3]. However, AUC is known to be insensitive to class imbalance and does not reflect predictive utility at specific decision thresholds. To address this limitation, I adopted the Matthews Correlation Coefficient (MCC), which incorporates all four cells of the confusion matrix and is regarded as one of the most informative metrics for imbalanced binary classification [4]. MCC enabled a search for an optimal classification threshold and provided a more realistic assessment of the model’s ability to distinguish deaths from survivors. This is deemed critical in a clinical setting where false negatives may carry significant cost (in this case, failing to identify a high-risk patient who might otherwise receive timely treatment to prevent death).

Regularisation via the elastic net was explored to evaluate whether shrinkage or variable selection could improve robustness. However, because regularisation penalties are scale-dependent, normalisation of continuous predictors was mathematically necessary. Without standardisation, predictors measured on larger scales (e.g. CPK levels in the thousands) would incur disproportionately large penalties and therefore be shrunk more aggressively than those measured on smaller scales (e.g. age) [5]. By centering and scaling the predictors, with a mean of 0 and standard deviation of 1, each coefficient became comparable on a standardised metric.

Despite the theoretical appeal of the elastic net, it did not outperform the weighted logistic regression model. This was likely due to the dataset’s relatively modest dimensionality ($n = 299$) and lack of strong multicollinearity. Ultimately, the spline-augmented logistic regression model was concluded to be the most appropriate model. It was flexible enough to capture non-linear relationships while retaining explanatory power. From a clinical standpoint, this is advantageous; risk functions that correctly capture curvature (such as the long-established non-linear association between renal impairment and mortality) provide more calibrated risk estimates, which in turn better support prognostic decision-making.

Nevertheless, several limitations warrant reflection. First, although the spline model improved calibration, the gain was small and moderate miscalibration persisted at higher predicted probabilities. Second, the evaluation was conducted using a single train-test split, which limits the stability of the performance estimates. In a clinical setting, external validity is paramount and, ideally, the model would undergo k -fold cross-validation, temporal validation or geographical validation using data from different hospitals (or populations) [6]. Without such validation, the model risks overfitting to the specific characteristics of the data set and may not generalise to new patients.

Another limitation relates to sample size and outcome imbalance. Although weighted logistic regression mitigates some of the issues caused by rarer outcomes, the effective sample size (particularly in the high-risk tail) remains small. This influenced the instability of measures such as MCC, where division by zero required numerical safeguards. Moreover, spline models, while flexible, can overfit when degrees of freedom are not carefully chosen. Although I used a conservative choice of $df = 3$, the model could still be sensitive to noise, especially due to sparsity in the predictors.

If the objective were to ‘win’ a competition, I would alter the methodological direction greatly. Rather than prioritising interpretability, I would consider gradient-boosted trees or random forests, as highlighted in Chicco and Jurman’s work on this data set. These methods naturally capture complex interactions and non-linearities without requiring manual feature engineering. I could also employ repeated cross-validation,

Bayesian hyperparameter search and, potentially, ensemble multiple learners to maximise discrimination. However, these gains would come at the cost of interpretability, which can be undesirable in a clinical context. Another avenue for future exploration would be to reinstate the `time` variable and perform a survival analysis, which aligns more closely with the way this data set was originally curated. This would provide a more appropriate modelling framework for time-to-event outcomes. Furthermore, the model must not only perform well statistically but also satisfy broader requirements: fairness across (sub)populations, robustness to missing data (common in health data) and clear interpretability to clinicians. Moreover, clinical prediction models must be externally validated and updated over time as population characteristics change. AUC or MCC alone are insufficient; calibration, decision-curve analysis and impact analyses would be required to fully justify deployment.

In summary, the strategy I developed balanced interpretability and performance evaluation appropriate for the data characteristics. The exploration of non-linear extensions and careful consideration of metrics under class imbalance reflect a thoughtful approach to predictive modelling. Future work should focus on areas such as external validation and incorporation of more flexible models when appropriate. Rigorous assessment of clinical utility is needed to strengthen the model’s readiness for real-world assessment.

References

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